

Lab #7

PIC18F Interrupt I/O using Assembly and C
ECE 3301L Introduction to Microcontrollers Laboratory

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Videos:

Prelab (part a):

Explanation of code:

https://youtu.be/cF_SUnyCHJA?si=5t844pAYSDpkspZ3

Physical circuit:

<https://youtu.be/SkFk5b-A7Rc?si=cxigEvCCm2ZP2HmO>

Prelab (part b):

Explanation of code:

<https://youtu.be/SjluJhsYwbU?si=RsUg4c7iWHYoDACI>

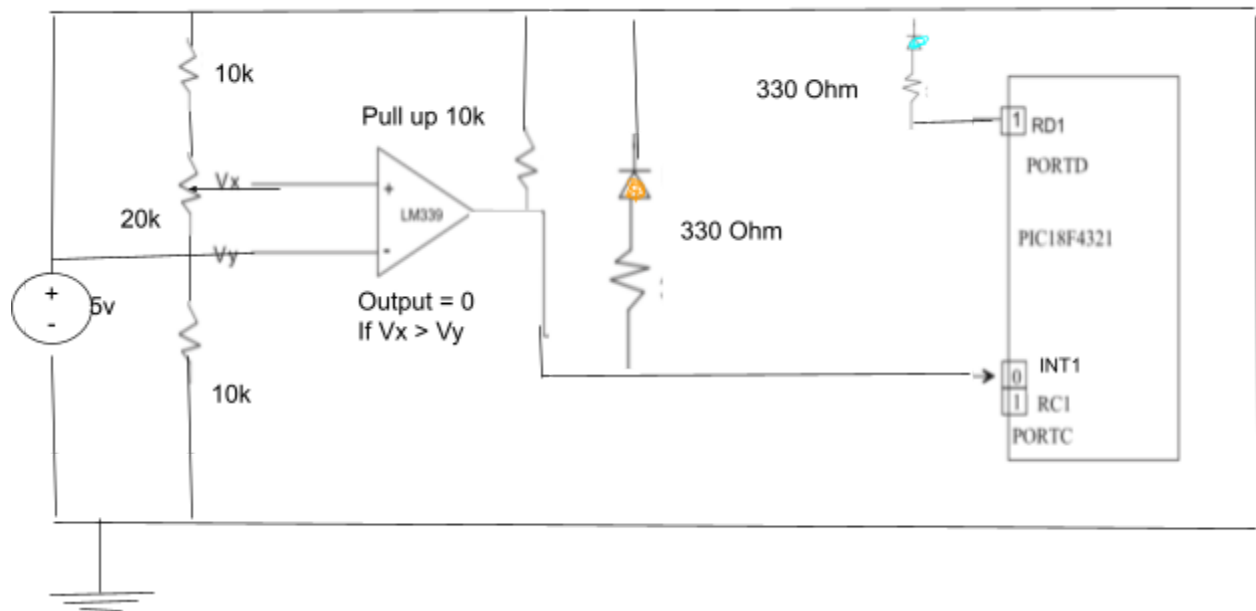
Physical circuit:

<https://youtu.be/X6M3K-IYV8Y?si=6oFRj08s5yn69FJI>

Objective: The purpose of this lab is to illustrate the concept of PIC18F4321 interrupt I/O.

PRELAB:

prelab i)



IMG 1. The schematic of the circuit

The schematic of the circuit has been slightly modified so that only one power source is required and that we can see the results of the comparator before it is processed as input by the Pic18F. The 5v serves as v_y , and is also the power for pins 11 and 32 of the PIC18F, and pin 3 of the LM399n. Everything is connected to the same ground. In this manner, only one voltage source needs to be used to have two different voltage inputs to compare as well as to provide power to the Pic and comparator. The voltage source v_x is adjusted using a 20k potentiometer, and when $v_x > v_y$ the comparator outputs a 0. The 0 turns on the LEDS, as it connects the LEDs to ground. The orange LED is directly from the output of the comparator- if $v_x > v_y$, then the comparator has an output of 0, connecting the orange LED to ground and turning it on. The blue LED is the output of the Pic18F- if bit 1 of port D is 0, then the LED will turn on. Using two LEDs allows us to compare the output results of the comparator and the Pic18F. Note that the PIC was pre-programmed before going into the circuit, so in-circuit connections to the pickit are unnecessary.

Instructions

a) Repeat LAB 7 using Interrupt I/O by connecting the comparator output to INT1. Note that RB1 is also multiplexed with INT1. Write the main program at 0x80 and the interrupt service routine at 0x150 in PIC18F assembly language. The main program will configure the I/O ports, enable interrupt INT1, initialize STKPTR to 0x10, turn the LED OFF, and then wait for interrupt. The interrupt service routine will turn the LED ON and return to the main program at the appropriate location so that the LED is turned ON continuously until the next interrupt.

(b) Repeat (a) using C except initialization of STKPTR is not required and also starting addresses for the main and the service routine do not need to be assigned. As shown in the above figure, the PIC18F4321 is required to turn on an LED connected to bit 1 of PORTD if the comparator voltage $V_x > V_y$; otherwise, the LED will be turned off.

Essentially, in the previous lab 6 we used polling (constantly checking) to monitor the comparator output and update the state of the LED. In this lab, we are doing the same thing but with interrupts. Arguably, this method (while more complicated to implement in our software) is better, because it lets the Pic18F “wait” until a comparator output change. Contrary to lab 6, instead of checking the comparator in a loop we will trigger an interrupt using INT1 (which is connected physically to the LM339n output pin). Since RB1 is the same pin as INT1, we need to make sure that RB1 is set as an input. We are writing this prelab code in assembly first for part A like previous labs. The main memory address should be at 0x80. The interrupt service routine (ISR) should be placed at memory location 0x150. For part B, we are writing our code in C. For part B, initialization of STKPTR is not required (so no need to write `movlw 0x10` or initialize STKPTR), and no need to manually assign main to location 0x80 and ISR to 0x150 (the compiler does this automatically). Bit one of port B is still an input (and output of the comparator), bit 1 of port D is still output going to the LED, INIT1 is enabled, and we are using an interrupt function in C.

Equipment, Software, and Components required:

- Microchip’s MPLAB assembler/debugger and C compiler
- Parts’ List
 - a) PicKit3 and PIC18F4321 from Microchip
 - b) Breadboard
 - c) Resistor (330 ohm) and LED
 - d) Resistors (1K and 10K), and Push button for RESET circuit
 - e) LM339 Comparator
 - f) Power Supply
 - g) Wires and Clip leads

Description (corresponding topics covered in the textbook)

Example 9.2, Appendix F, Appendix G, Appendix H

Prerequisites

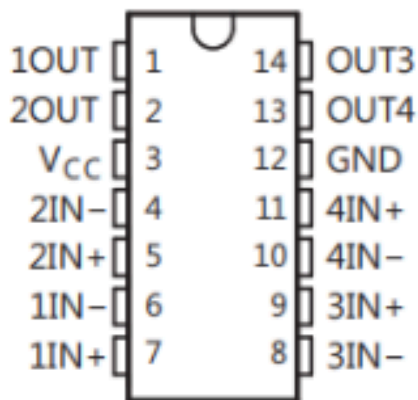
Sections 9.2

Procedure

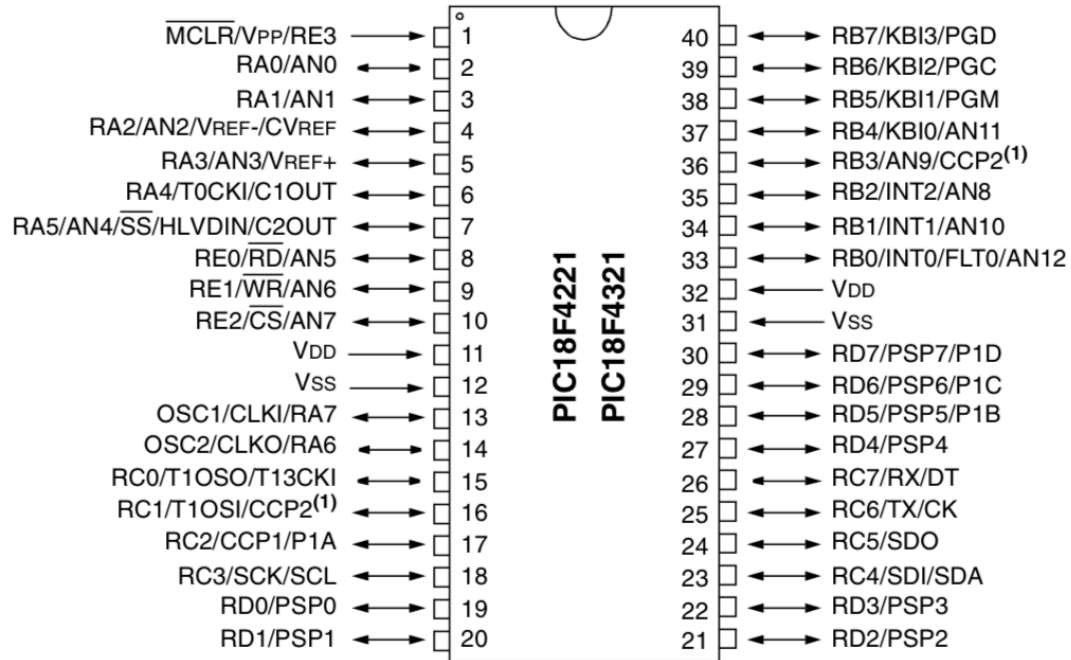
- Assemble the PIC18F assembly language program and compile the C-program using the MPLAB.
- Download the assembled or the compiled program into the PIC18F4321 on the breadboard from your Personal Computer or Laptop using the PICKit3 and MPLAB following the steps provided in Appendix H of the book.
- Use the default clock of the PIC18F4321 and connect the appropriate RESET circuit to the PIC18F4321 pin.
- Connect the hardware and demonstrate the lab as a PIC18F4321-based stand-alone system.

Resources for hardware

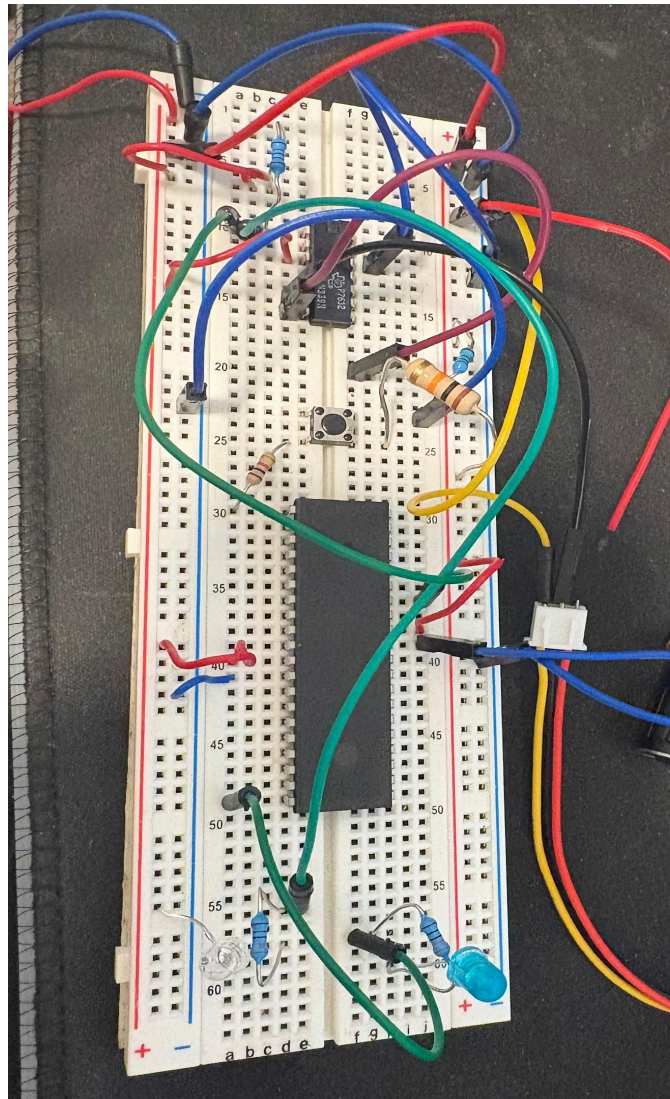
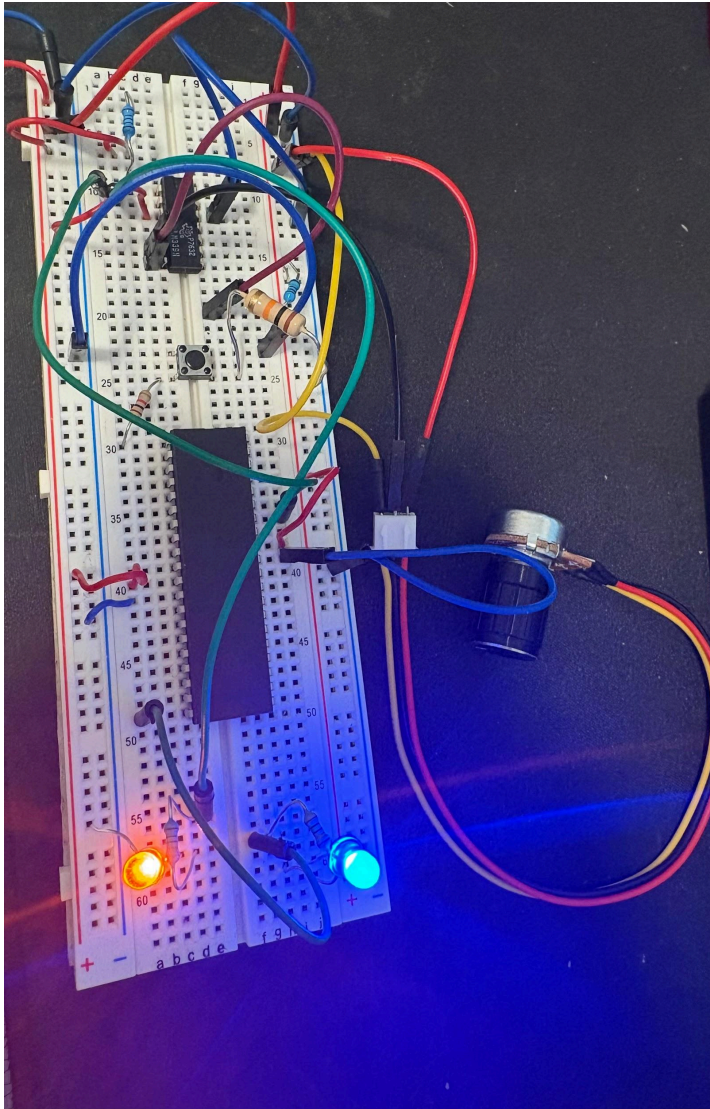
The logistics of the hardware were explained previously. Below are resources used to create the circuit.



IMG 2. Layout of the LM339n comparator, from [Ovega.com](https://www.ovaga.com) (fully cited at the end of the report).

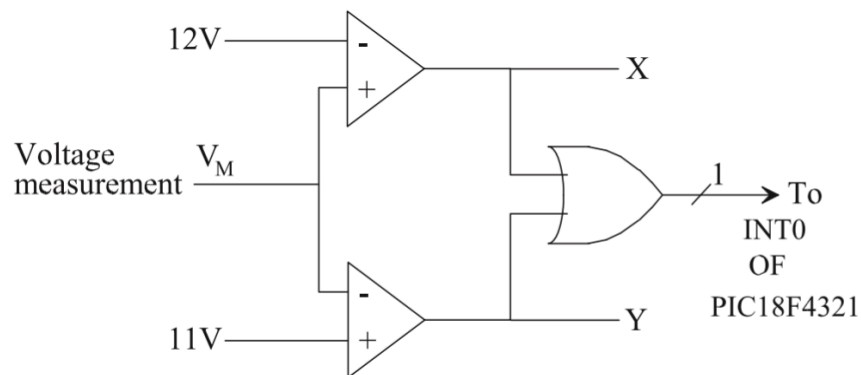


IMG 3. Layout of the PIC18F4321, from microchip.com. More information is at the end of this report.



IMG 4 and 5. Physical circuit from the prelab

Postlab



In the above figure, if $V_M > 12\text{ V}$, turn an LED ON connected at bit 3 of PORTA. If $V_M < 11\text{ V}$, turn the LED OFF. Using ports, registers, and memory locations as needed and INT0 interrupt:

(a) Draw a neat block diagram showing the PIC18F4321 microcontroller and the connections to ports in the diagram shown in Figure P9.11.

- *Microchip*, ww1.microchip.com/downloads/en/DeviceDoc/39689b.pdf. Accessed July 2025.

www.ovaga.com, Ovaga |. "LM339N IC Datasheet: Circuit, Pinout, and Uses." *Ovaga*, www.ovaga.com/blog/transistor/lm339n-ic-datasheet-circuit-pinout-and-uses?srsId=AfmBOorbA8F4Gfqacw8ExYRYTT2PoRaCAYdzCk6Rgl7YlqrG4kG3MSk. Accessed 30 July 2025.

- Www.jotrin.com, Jotrin |. "LM339N Comparator Datasheet, Pinout, and Circuit." *Jotrin Electronics*, www.jotrin.com/technology/details/lm339n-datasheet-circuit-and-pinout. Accessed July 2025.